

Executive summary. The most defensible strategy is to place oxygen half-saturation priors near recognized tumor hypoxia thresholds, center the Richards shape parameter near the standard logistic ($v = 1$), assign a day-scale prior to oxygen relaxation, and treat $\epsilon_{\log N}$ as a biologically interpretable floor tied to the smallest meaningful burden scale rather than as an arbitrary purely numerical constant.

Unit conversion used throughout. For practical prior specification, 1% O₂ is treated as approximately 7.6–8 mmHg. Common anchors are 5% $\approx$ 38 mmHg, 2% $\approx$ 15 mmHg, and 1% $\approx$ 8 mmHg. Tumor median pO₂ is often around 10 mmHg (about 1.3–1.4% O₂), providing a reasonable biological reference scale for latent-oxygen priors.

1. k_o : oxygen half-saturation constant in the proliferation response

Definition. Oxygen level at which the proliferation-driving oxygen response is half-maximal.

Evidence summary. Tumor proliferation is often maintained under modest hypoxia but becomes progressively suppressed under more severe oxygen deprivation. A defensible prior should therefore sit near the physiological-to-pathological hypoxia boundary rather than near normoxia. In practice, centering the prior around roughly 1–3% O₂ keeps the model compatible with the fact that many untreated tumors spend substantial time in the low-percent oxygen range while still growing.

Prior candidates.

- Conservative prior: $k_o \sim \text{LogNormal}(\log(1.8), 0.35)$ in % O₂, truncated to [0.5, 5].
- Broad prior: $k_o \sim \text{LogNormal}(\log(2.0), 0.70)$ in % O₂, truncated to [0.25, 8].
- Equivalent center in mmHg: about 14–15 mmHg.

Sensitivity checks. Refit with prior medians fixed at 1%, 2%, and 3% O₂. If posterior growth predictions change strongly across that band, then the oxygen block is weakly identified and the dependence on prior choice should be reported explicitly.

2. $k_{\{o, \text{mis}\}}$: oxygen half-saturation constant in the missegregation/CIN-stress response

Definition. Oxygen level at which the hypoxia-linked missegregation or chromosomal-instability stress response reaches half of its maximum.

Evidence summary. Direct oxygen-to-missegregation parameterizations are rare, so the defensible anchor comes from the broader hypoxia, reoxygenation, ROS, and genomic-instability literature. Because pimonidazole-relevant hypoxia begins near 10 mmHg and stronger stress signatures emerge at still lower oxygen, it is reasonable to place this prior at or slightly below the proliferation half-saturation prior.

Prior candidates.

- Conservative prior: $k_{\{o, \text{mis}\}} \sim \text{LogNormal}(\log(1.0), 0.40)$ in % O₂, truncated to [0.2, 3].

- Broad prior: $k_{\{o,mis\}} \sim \text{LogNormal}(\log(0.8), 0.80)$ in % O₂, truncated to [0.1, 5].
- Equivalent center in mmHg: about 6–8 mmHg.

Sensitivity checks. Compare one calibration with the structural constraint $k_{\{o,mis\}} < k_o$ against another with no ordering constraint. If the model only fits when $k_{\{o,mis\}}$ is much larger than k_o , the term is probably acting as a generic latent-stress knob rather than an oxygen-stress threshold.

3. v_{O_2} : shape parameter in the general logistic oxygen-vs-burden curve

Definition. Dimensionless Richards/general-logistic shape parameter controlling asymmetry of the burden-to-oxygen target curve.

Evidence summary. The standard logistic is recovered when $v = 1$. Moderate departures from 1 capture earlier- or later-than-logistic transitions without introducing implausibly sharp curve shapes. Because this parameter is hard to identify when oxygen is latent, the most defensible prior is centered at 1 with only moderate spread unless data strongly support asymmetry.

Prior candidates.

- Conservative prior: $v_{O_2} \sim \text{LogNormal}(\log(1), 0.35)$, truncated to [0.4, 2.5].
- Broad prior: $v_{O_2} \sim \text{LogNormal}(\log(1), 0.80)$, truncated to [0.2, 5].

Sensitivity checks. Compare the full model against a reduced model with v_{O_2} fixed at 1. If fit quality changes minimally, fixing $v_{O_2} = 1$ improves interpretability and reduces prior sensitivity.

4. τ_{O_2} : oxygen relaxation time constant

Definition. Time constant, in days, describing how quickly tumor oxygen relaxes toward the burden-determined target.

Evidence summary. This parameter should represent the bulk adaptation timescale of tumor oxygenation, not the near-instantaneous chemistry of oxygen sensing. Acute and cycling hypoxia can fluctuate over minutes to hours, but growth-linked changes in average tumor oxygenation and hypoxic fraction in untreated tumors are typically observed across days. A prior on the order of one to a few days is therefore the most defensible grant-ready choice.

Prior candidates.

- Conservative prior: $\tau_{O_2} \sim \text{LogNormal}(\log(2), 0.45)$ days, truncated to [0.25, 7].
- Broad prior: $\tau_{O_2} \sim \text{LogNormal}(\log(2), 0.90)$ days, truncated to [0.1, 14].

Sensitivity checks. Repeat calibration with τ_{O_2} fixed at 0.5, 2, and 5 days. If predictions are robust, retain the conservative prior. If predictions are highly sensitive, then the lag structure is functionally important despite oxygen being unobserved.

5. $\epsilon_{\log N}$: additive offset inside $\log_{10}(N + \epsilon_{\log N})$

Definition. Offset, in cells, used inside the logarithm to prevent pathological behavior at very small burdens and to define a biologically interpretable floor.

Evidence summary. This parameter is primarily regularizing, but it should still be tied to a meaningful burden scale. Small imaging-detectable lesions and microtumor or spheroid-scale burdens provide a principled anchor. Values in the 10^2 to 10^4 cell range are usually large enough to stabilize early-time dynamics while remaining negligible once tumors become macroscopic.

Prior candidates.

- Conservative prior: fix $\epsilon_{\log N} = 10^3$ cells.
- Broad prior: $\log_{10}(\epsilon_{\log N}) \sim \text{Uniform}(2, 4)$, corresponding to $\epsilon_{\log N}$ in $[10^2, 10^4]$ cells.

Sensitivity checks. Refit with $\epsilon_{\log N} = 10^2, 10^3$, and 10^4 . If early dynamics or inferred oxygen trajectories shift materially, then $\epsilon_{\log N}$ is biologically active in the model and should not be described as purely numerical.

Grant-facing recommendation

Recommended prior pack. For a grant or methods appendix, the strongest default package is: $k_o \sim \text{LogNormal}(\log(1.8), 0.35) \% \text{O}_2$; $k_{\{o, \text{mis}\}} \sim \text{LogNormal}(\log(1.0), 0.40) \% \text{O}_2$; $v_{\text{O}_2} \sim \text{LogNormal}(\log(1), 0.35)$; $\tau_{\text{O}_2} \sim \text{LogNormal}(\log(2), 0.45)$ days; and either $\epsilon_{\log N} = 10^3$ cells or $\log_{10}(\epsilon_{\log N}) \sim \text{Uniform}(2, 4)$. This package is biologically grounded, conservative in shape assumptions, and transparent about where the latent-oxygen component is informed by external evidence rather than directly by the calibration data.

Evidence table

Parameter / anchor	Model system	Scale	Functional form / modality	Extracted value(s)	Notes	DOI / PMID
Conversion anchors	McKeown 2014 review	mmHg $\leftrightarrow$ % O ₂	Review / oxygen-scale synthesis	5% $\approx$ 38 mmHg; 2% $\approx$ 15 mmHg; 1% $\approx$ 8 mmHg	Convenient conversion anchors for prior specification	PMID 24588669
Tumor oxygen regime	McKeown 2014 review	mmHg / % O ₂	Review of solid tumors	Median tumor pO ₂ around 10.27 mmHg ($\sim$ 1.4% O ₂)	Supports low-percent oxygen centering for latent-oxygen priors	PMID 24588669
Pimonidazole threshold	Koch 2008	mmHg	Hypoxia marker kinetics	Binding reported as zero above 10 mmHg, increasing below	Useful anchor for oxygen-stress thresholding	PMID 18494550
Severe hypoxia and proliferation suppression	Wadsworth et al. 2022	% O ₂	Experimental review / xenograft context	0.5% O ₂ strongly suppresses proliferation; 1% O ₂ can still permit some proliferation	Supports k_o below normoxic but above extreme-anoxia levels	PMID 35508768
Oxygen-limited proliferation precedent	Casciari et al. 1992	mM O ₂	Empirical growth-rate relation	Growth cessation only under very severe oxygen deprivation	Argues against centering k_o too high	DOI 10.1002/jcp.1041510220
Hypoxia/CIN link	Kondoh et al. 2013	Hypoxia / reoxygenation	Aneuploidy assay	Aneuploidy induced within 7 days under hypoxia or hypoxia-reoxygenation	Best direct biological support for $k_{\{o, \text{mis}\}}$	DOI 10.1371/journal.pone.0080349

Cycling vs chronic hypoxia timescale	Luoto et al. 2013; Saxena & Jolly 2019	Conceptual	Reviews	Cycling hypoxia on minutes-to-hours scale; chronic hypoxia >24 h	Supports day-scale bulk τ_{O_2} rather than minute-scale τ_{O_2}	PMID 24152759; PMID 31382593
Growth-linked oxygen change	Bratasz et al. 2007; Ilangoan et al. 2004	mmHg	Longitudinal EPR imaging	Tumor pO ₂ decreased with growth and reached hypoxic levels within days	Supports τ_{O_2} on the order of days	PMID 17457861; PMID 15334586
Richards asymmetry parameter	Tj�rve & Tj�rve 2010; Dawson et al. 2012	Unitless	Richards / asymmetric logistic	$v = 1$ gives logistic symmetry; moderate deviations give asymmetry	Supports centering v_{O_2} at 1	PMID 20831877; PMID 22210403
Small-burden biological floor	Thorsen et al. 2014; Puaux et al. 2011	Cells / lesion size	Imaging detection limits	Sub-mm lesions correspond roughly to 10^2 – 10^3 cells depending on assumptions	Supports $\epsilon_{\log N}$ in the 10^2 – 10^4 range	PMID 24349320; PMID 22163373

Compact prior pack

- $k_o \sim \text{LogNormal}(\log(1.8), 0.35)$ in % O₂, truncated to [0.5, 5].
- $k_{\{o, \text{mis}\}} \sim \text{LogNormal}(\log(1.0), 0.40)$ in % O₂, truncated to [0.2, 3].
- $v_{O_2} \sim \text{LogNormal}(\log(1), 0.35)$, truncated to [0.4, 2.5].
- $\tau_{O_2} \sim \text{LogNormal}(\log(2), 0.45)$ days, truncated to [0.25, 7].
- Either fix $\epsilon_{\log N} = 10^3$ cells, or use $\log_{10}(\epsilon_{\log N}) \sim \text{Uniform}(2, 4)$.

Key references

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